

ODE

Assignment

Assignment-3

1.1 (a) We have, $y' = y^{1/2}$ with $y(0) = 1$.
Let $f(y) = y^{1/2}$ which is clearly continuous in a neighbourhood of the initial value $y(0) = 1$ (eg. on the interval $(0, \infty)$). And the derivative $f'(y) = \frac{1}{2\sqrt{y}}$ which is continuous and bounded in any closed interval around $y(0) = 1$. So, the function is locally Lipschitz.

By the Picard-Lindelöf theorem, a unique solution exists around $t=0$.

Explicitly solving ODE, we get

$$\int \frac{dy}{y^{1/2}} = \int dt \Rightarrow \int y^{-1/2} dy = \int dt \Rightarrow 2\sqrt{y} = t + c$$

By using initial condition $y(0) = 1$, we get

$$\Rightarrow 2 = 0 + c \Rightarrow \boxed{c=2}$$

The solution is, $y(t) = \left(\frac{t+2}{2}\right)^2 = \frac{(t+2)^2}{4}$.

For this solution to be valid it must satisfy the ODE $y' = \sqrt{y}$.

$$\therefore y' = \frac{t+2}{2}; \quad \sqrt{y} = \frac{|t+2|}{2}$$

Now as per the ODE, $\frac{t+2}{2} = \frac{|t+2|}{2}$

$\Rightarrow t+2 = |t+2|$ for this to be valid $t+2 \geq 0$

i.e. $t \geq -2$. The solution doesn't blow up,

so the maximal interval of existence is $[-2, \infty)$

(b) We have, $y' = y^{1/3}$. Let $f(y) = y^{1/3}$ which is continuous near the initial value $y(0) = 1$.

Also its derivative $f'(y) = \frac{1}{3}y^{-2/3}$ which is continuous and bounded in a neighbourhood of $y(0) = 1$. So, the function is locally Lipschitz around $t=0$. Therefore a unique solution exists locally.

Explicitly solving the ODE, we get

$$\frac{dy}{dt} = y^{1/3} \Rightarrow \int y^{-1/3} dy = \int dt \Rightarrow \frac{3}{2}y^{2/3} = t + c$$

using the initial value $y(0) = 1$, we get $\boxed{c=3/2}$

The solution is $\frac{3}{2}y^{2/3} = t + 3/2 \Rightarrow y^{2/3} = \frac{2t+3}{3}$

which gives $y(t) = \left(\frac{2t+3}{3}\right)^{3/2}$. This solution is defined real valued solution as long as $\frac{2t+3}{3} \geq 0$

$\Rightarrow t \geq -3/2$. The solution does not blow up

so, the maximal interval of existence is $[-3/2, \infty)$

1.2 (a) Given a C^1 function $\sigma(x)$ with $\sigma(x) > 0$ and $|\sigma'(x)| \leq K\sigma(x)$. This is equivalent to $-K\sigma(x) \leq \sigma'(x) \leq K\sigma(x)$. Assuming $\sigma(x) > 0$ we can divide by it,

$$-K \leq \frac{\sigma'(x)}{\sigma(x)} \leq K \Rightarrow -K \leq (\ln \sigma(x))' \leq K$$

Case I: $x > a$.

Integrating from a to x with respect to t , we get

$$-\int_a^x K dt \leq \int_a^x (\ln \sigma(t))' dt \leq \int_a^x K dt$$

Exponentiating all parts give:

$$= e^{-K(x-a)} \leq \frac{\sigma(x)}{\sigma(a)} \leq e^{K(x-a)}$$

$$\Rightarrow \sigma(a)e^{-K(x-a)} \leq \sigma(x) \leq \sigma(a)e^{K(x-a)}$$

$$\Rightarrow \sigma(a)e^{-K|x-a|} \leq \sigma(x) \leq \sigma(a)e^{K|x-a|}$$

Case II: $x < a$

Integrating from x to a , we get

$$-\int_x^a K dt \leq \int_x^a (\ln \sigma(t))' dt \leq \int_x^a K dt$$

$$\Rightarrow -K(a-x) \leq \ln \sigma(a) - \ln \sigma(x) \leq K(a-x)$$

Exponentiating both sides, we get

$$\Rightarrow e^{-K(a-x)} \leq \frac{\sigma(a)}{\sigma(x)} \leq e^{K(a-x)}$$

Rearranging for $\sigma(x)$ yields:

$$\sigma(a)e^{-K(a-x)} \leq \sigma(x) \leq e^{K(a-x)} \sigma(a)$$

$$\Rightarrow \sigma(a)e^{-K|x-a|} \leq \sigma(x) \leq \sigma(a)e^{K|x-a|}$$

Thus, the inequality holds for all $x \in \mathbb{R}$

(b) Proof of uniqueness for the IVP

Let $y_1(x)$ and $y_2(x)$ be two solutions to the IVP: $y' = f(x, y)$, $y(x_0) = y_0$. Let's define the function $\sigma(x) = (y_1(x) - y_2(x))^2 + \varepsilon$ for any arbitrary $\varepsilon > 0$. This ensures $\sigma(x) > 0$

Taking the derivative, we get

$$\sigma'(x) = 2(y_1(x) - y_2(x)) \cdot (y_1'(x) - y_2'(x))$$

$$\Rightarrow \sigma'(x) = 2(y_1(x) - y_2(x)) \cdot (f(x, y_1) - f(x, y_2))$$

Using the given global Lipschitz condition

$$|f(x, y_1) - f(x, y_2)| \leq K|y_1 - y_2|$$

$$|\delta'(x)| = 2|y_1 - y_2| |f(x, y_1) - f(x, y_2)| \leq 2K|y_1 - y_2|^2$$

$\therefore (y_1 - y_2)^2 = \delta(x) - \varepsilon < \delta(x)$, we have

$|\delta'(x)| \leq 2K\delta(x)$ which matches condition of part (a) with constant $2K$.

Applying the result of part (a), using $a = x_0$ we get

$$\delta(x_0)e^{-2K|x-x_0|} \leq \delta(x) \leq \delta(x_0)e^{2K|x-x_0|} \quad (i)$$

At the initial point x_0 , we have

$$y_1(x_0) = y_2(x_0) = y_0$$

$$\therefore \delta(x_0) = (y_1(x_0) - y_2(x_0))^2 + \varepsilon = \varepsilon \quad (ii)$$

Using eq. (ii) in the inequality of eq. (i), we get

$$\varepsilon e^{-2K|x-x_0|} \leq (y_1(x) - y_2(x))^2 + \varepsilon \leq \varepsilon e^{2K|x-x_0|}$$

This must hold for all $\varepsilon > 0$. Taking the limit as $\varepsilon \rightarrow 0^+$:

$$0 \leq (y_1(x) - y_2(x))^2 \leq 0. \text{ This implies that } (y_1(x) - y_2(x))^2 = 0 \Rightarrow y_1(x) = y_2(x) \text{ for all } x.$$

Thus, the solution is unique. The existence of global solution is also guaranteed by global Lipschitz condition.

(c) The condition is $|f(x, y_1) - f(x, y_2)| \leq K|y_1 - y_2| \cdot |\log|y_1 - y_2||$. This is a case for the Osgood's uniqueness theorem. The theorem guarantees a unique solution if the integral $\int_0^\delta \frac{du}{g(u)}$ diverges, where $g(u)$ is the function on the R.H.S. Here, $g(u) = Ku|\log u|$. Let's check the integral for small $\delta < 1$:

$$\int_0^\delta \frac{du}{Ku|\log u|} = -\frac{1}{K} \int_0^\delta \frac{du}{u \log u} \quad (\because \log u < 0)$$

Let $v = -\log u$, so $dv = -\frac{du}{u}$. As $u \rightarrow 0^+$, $v \rightarrow +\infty$

$$\therefore -\frac{1}{K} \int_{-\log \delta}^{+\infty} \frac{dv}{v} = -\frac{1}{K} [\log v]_{-\log \delta}^{+\infty} \rightarrow +\infty$$

This integral diverges to ∞ . Therefore, by Osgood's criterion, the solution to the IVP is unique. Since, the condition also implies continuity in y . By Peano's theorem, it guarantees that a solution exists.

1.3. The given ODE, $y' = 3y^{2/3}$ with $y(0) = 0$. Let $f(y) = 3y^{2/3}$ its derivative $f'(y) = 2y^{-1/3}$. As $y \rightarrow 0$, $|f'(y)| \rightarrow \infty$. The derivative is unbounded near $y = 0$, so the function is not Lipschitz continuous at the initial value.

The function $f(y)$ is continuous everywhere. By Peano's theorem, a solution exists.

Explicitly solving ODE, we get

$$\frac{dy}{dt} = 3y^{2/3} \Rightarrow \int y^{-2/3} dy = \int 3 dt \Rightarrow 3y^{1/3} = 3t + c$$

Using $y(0) = 0$, we get $c = 0$. The solution is $y = t^3$.

This shows that Lipschitz condition is not necessary for existence.

The solution is not unique because $y(t) = 0$ for all t is also a valid solution as it satisfies the ODE. Since, we find two distinct solutions $y_1(t) = t^3$ and $y_2(t) = 0$ satisfying the same initial conditions, the solution is not unique.

If $y(0) = 1$ then in a neighbourhood of $y(0) = 1$ the derivative $f'(y) = 2y^{-1/3}$ is bounded near $y(0) = 1$. Therefore, $f(y)$ is locally Lipschitz near $y(0) = 1$. By the applying of Picard-Lindelöf theorem it guarantees a unique solution which exists locally. The solution is $y(t) = (t+1)^3$.

1.4. The given ODE, $y' = 1 + y^{2/3}$, with $y(0) = 0$. Let $f(y) = 1 + y^{2/3}$ and its derivative $f'(y) = \frac{2}{3}y^{-1/3}$. As $y \rightarrow 0$, the derivative is unbounded. Therefore, the function is not Lipschitz continuous at the initial value $y = 0$. Since, $f(y) = 1 + y^{2/3}$ is continuous, a solution exists by Peano's theorem.

Now any solution starting at $y = 0$ must immediately move away from 0 (since, $y'(0) = 1 > 0$). Once, $y(t) \neq 0$, the solution enters the region where $f(y)$ is Lipschitz continuous, which enforces uniqueness from that point on. Because the solution is forced to leave the non-Lipschitz point $y = 0$ immediately, different solution paths cannot branch off. This demonstrates that the Lipschitz condition is not necessary for uniqueness.

The unique solution implicitly is given by $3(y^{1/3} - \arctan(y^{1/3})) = t$.

1.5 The given ODE, $y'(t) = \frac{e^{-y^2} \sin^2 t}{1+t^2}$ with $y(0)=1$.
 Let $f(t, y) = \frac{e^{-y^2} \sin^2 t}{1+t^2}$.

Differentiate $f(t, y)$ with respect to y , we get
 $\left| \frac{\partial f(t, y)}{\partial y} \right| = \left| \frac{2y e^{-y^2} \sin^2 t}{1+t^2} \right| \leq 2|y| e^{-y^2} \leq M$

where M is the maximum value of $2y e^{-y^2}$.
 Therefore, $\frac{\partial f}{\partial y}$ is globally bounded, so f is globally Lipschitz in y . By Picard-Lindelöf we get existence and uniqueness of a global solution. Since f is continuous in t and globally Lipschitz in y and moreover $|f(t, y)| \leq \frac{e^{-y^2}}{1+t^2} \leq \frac{1}{1+t^2}$ (uniformly in y), the solution cannot blow up, it extends $\forall t \in \mathbb{R}$. Hence, the maximal interval is $\mathbb{R}$.

1.6 Let $x' = f(x)$ where $f(x) = \begin{cases} x \sin(1/x), & x \neq 0 \\ 0, & x = 0 \end{cases}$.
 The function is continuous at $x=0$ because $\lim_{x \rightarrow 0} x \sin(1/x) = 0 = f(0)$. Since, $f(x)$ is continuous everywhere, a solution exists, by Peano's Theorem.

The derivative $f'(x) = \sin(1/x) - \frac{1}{x} \cos(1/x)$ which is unbounded as $x \rightarrow 0$, so the function is not Lipschitz at $x=0$. However, the solution is unique. As if $x(t)$ is any solution with $x(0)=0$, is of the form $x(t) = \int_0^t f(x(s)) ds$

$$\text{so, } |x(t)| = \left| \int_0^t f(x(s)) ds \right| \leq \int_0^t |f(x(s))| ds \leq \int_0^t |x(s)| ds$$

$$\therefore |x(t)| \leq \int_0^t |x(s)| ds$$

Apply Gronwall this yields $x(t) \equiv 0$ $\forall t$ in the interval of existence. Therefore the only solution with $x(0)=0$ is the trivial solution $x(t)=0$.

1.7 Let $f(t, y) = p(t) \cos y + q(t) \sin y$. The function $p(t)$ and $q(t)$ are continuous on $\mathbb{R}$.

The function $f(t, y)$ is continuous on $\mathbb{R} \times \mathbb{R}$. Let's check the Lipschitz condition in y :

$$\left| \frac{\partial f}{\partial y} \right| = |-p(t) \sin y + q(t) \cos y| \leq |p(t)| + |q(t)|$$

on any compact interval of $[-T, T]$, the continuous function $|p(t)|$ and $|q(t)|$ are bounded by some constant M_T . Thus $\left| \frac{\partial f}{\partial y} \right| \leq M_T$,

which means $f(t, y)$ is locally Lipschitz in y . By the Picard-Lindelöf theorem, a unique solution $\phi(t)$ exists locally for any initial condition $\phi(t_0) = y_0$.

$$\therefore |y'(t)| \leq |p(t)| + |q(t)|$$

Integrating from t_0 to t , we get

$$\left| \int_{t_0}^t y'(s) ds \right| \leq \int_{t_0}^t |y'(s)| ds \leq \int_{t_0}^t (|p(s)| + |q(s)|) ds$$

$$\Rightarrow |y(t) - y(t_0)| \leq \int_{t_0}^t (|p(\tau)| + |q(\tau)|) d\tau$$

$\therefore p$ and q are continuous, the integral on the right is finite for any finite t .

This means $|y(t)|$ cannot go to infinity.

By the extension theorem, the unique solution ϕ must exist on $\mathbb{R}$. Therefore the maximal interval of existence is $\mathbb{R}$ i.e. $(-\infty, \infty)$.

